

Simulated Borehole Fracture during Non-reactive and Reactive Fluid Injection

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ABSTRACT: Optimizing carbon sequestration requires a better understanding of reservoir behavior during the injection of non-reactive and reactive fluids. This study adopts a discrete element model coupled with pore-scale multiphysics to investigate borehole pressurization and fracture development in low-permeability ultramafic rock such as peridotite. The numerical specimen is represented by a bonded granular assembly calibrated against experimental data. Fluid flow is simulated through a pore network that evolves with rock deformation and damage, while reactive injection accounts for mineral reactions that clog the pore space and induce solid volume changes. To simulate field operations, the simulations reproduce a sequence of injection, shut-in, and reopening. The results show rapid pressure buildup during constant-rate injection followed by a gradual pressure decline as fractures increase hydraulic connectivity. For non-reactive injection, changing the shut-in time can shift the timing of pressure dissipation and recovery. Reactive injection produces higher transient pressures and stronger post-restart fluctuations, indicating intermittent changes in permeability. Crack patterns further reveal that reactive injection promotes a longer, more branched fracture network and a wider damage zone compared with purely hydraulic stimulation. These findings highlight the importance of chemomechanical coupling effects for managing injectivity and permeability evolution during in-situ mineralization operations.

1 INTRODUCTION

Decarbonization is essential for mitigating ongoing ecosystem deterioration. Among carbon-removal options, one of the most promising approaches is mineral carbonation, which converts carbon dioxide into stable carbonate minerals such as calcite and magnesite. In-situ mineral carbonation in mafic and ultramafic rocks has already been tested at the borehole scale in Iceland and Oman, providing evidence of feasibility and effectiveness (Gíslason et al., 2018; Kelemen et al., 2021). Compared with other geological storage methods that rely mainly on structural or stratigraphic trapping beneath low-permeability caprock, mineral carbonation is less risky. Once carbon dioxide is converted to solid carbonates, buoyancy-driven migration and long-term leakage risks are significantly reduced. In practice, two major types of candidate rocks are considered: basalt and peridotite. Basalt targeted by the CarbFix project exhibits relatively high initial permeability ($\sim 7.7 \times 10^{-12} \text{ m}^2$) (Aradóttir et al., 2012; Callow et al., 2018). During injection, dissolution of host basalt near the well can help maintain injectivity, whereas carbonate

precipitation farther from the well will reduce permeability by clogging the pore space. In contrast, peridotite in the Oman drilling project has a much lower initial permeability ($\sim 10^{-18}$ to $\sim 10^{-21} \text{ m}^2$) than basalt (Kelemen et al., 2021). Enhancing the permeability is therefore a central challenge for ultramafic rock reservoirs. Two potential approaches are discussed here. The first approach is to promote fracturing through reaction-driven volume expansion. In fact, hydration and carbonation tend to cause an increase in solid volume, which, if accompanied by expansion-induced cracking, can open new flow paths and enhance permeability (Nisbet et al., 2024). Specifically, sufficiently strong eigenstrain gradients may induce cracking and thereby accelerate carbonation by improving fluid transport. Another approach is hydraulic fracturing, in which high-pressure fluid injection creates fractures and additional reactive surface area for storage. Despite the encouraging evidence, borehole-scale simulations that couple pressurization, reactions, and fracture evolution remain limited in the literature.

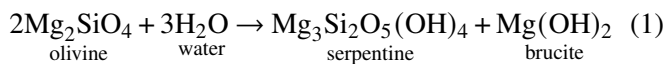
In order to investigate the chemomechanical coupling pro-

cess further, different numerical models have been proposed. These models can be classified into two main categories: continuum (Evans et al., 2020; Feng et al., 2025) and discrete (Zhang et al., 2019; Wu and Buscarnera, 2025). The continuum approach is generally easier to scale to field-scale and long-term simulations based on a homogenized approximation of the underlying mechanisms. To complement continuum approaches, discrete modeling has provided further insights into how reactions, stress redistribution, and damage localization jointly control permeability evolution. For example, the spring network model of Jamtveit et al. (2008) indicates that stress heterogeneity is a crucial driver of microfracture network development and the associated permeability increase. Shimizu and Okamoto (2016) further showed that the transition between fracturing and non-fracturing behavior is governed by the Damköhler number, highlighting the role of reaction–transport competition. In the context of volume-expansion reactions, Zhang et al. (2019) simulated porosity evolution and demonstrated that discrete predictions can qualitatively reproduce experimental measurements.

In this study, a discrete element model coupled with pore-scale multiphysics is used to simulate borehole pressurization and fracture development in low-permeability ultramafic rock. Non-reactive and reactive fluid injection are compared under an injection–shut-in–reopening sequence to quantify how mineral reactions interact with damage and hydraulic connectivity, thereby controlling transient pressure response and fracture network evolution during in-situ carbon storage operations.

2 MODEL SETUP AND CALIBRATION

In this study, both reactive and non-reactive fluid injections are examined. The non-reactive case represents purely hydraulic fracturing conditions. In the reactive case, this study isolates olivine hydration from competing reactions (Okamoto et al., 2011) because it predominantly drives the volume expansion that causes fracturing:



Here, the reaction is represented in a simplified stoichiometric form as

$$rW + A \rightarrow sB \quad (2)$$

where W denotes water and A denotes olivine. The product term B represents the combined reaction products (i.e.,

$\text{Mg}_3\text{Si}_2\text{O}_5(\text{OH})_4 \cdot \text{Mg}(\text{OH})_2$). Normalizing the stoichiometry to one mole of olivine yields $r = 3/2$ and $s = 1/2$, representing 0.5 moles of the combined solid products.

A two-dimensional numerical model is established as an idealized analogue of the in-situ tests. The model relies on a Discrete Element Method (DEM) enhanced with pore-scale multiphysics (Wu and Buscarnera, 2025). The numerical specimen contains 6,458 circular particles with radii ranging from 0.20 to 0.24 mm. The overall sample size is about 3 cm × 3 cm, providing a sufficiently large assembly to satisfy representative volume element (RVE) requirements (Wu et al., 2024). In our previous work, the coupled model was validated against the experimental results reported in Uno et al. (2022). The model setup and corresponding boundary conditions are shown in Fig. 1. The top and bottom boundaries are rigid, whereas the left and right boundaries are compliant (flexible membrane) to better accommodate rock deformation (volume expansion). The isotropic confining pressure is set to be 20 MPa. Fluid is injected at the borehole through a prescribed flux boundary during the injection and reopening stages, whereas the external specimen boundary remains impermeable. The rock deformation and fracturing processes are replicated using a standard DEM model based on bonded particles (Potyondy and Cundall, 2004), while a flow network model is adopted to simulate reactive transport (Fatt, 1956). To couple the two models, the pore network is constructed by placing its nodes at the center of the domains enclosed by loops of contacting particles. These network nodes are updated in conjunction with particle motion and the consequent rock volume change.

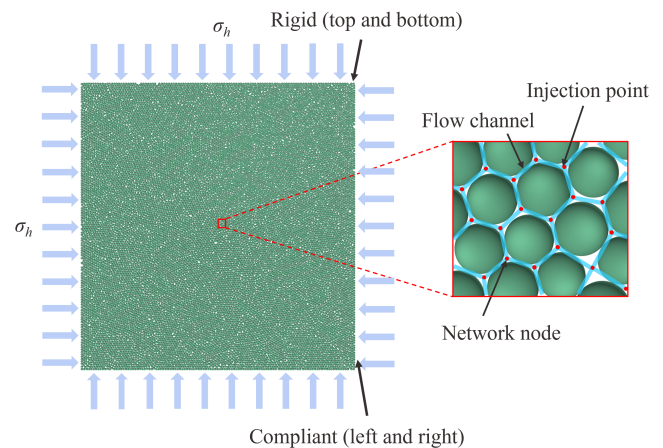


Fig. 1: Schematic of the numerical model used in the study.

Assuming Newtonian flow through narrow channels, the volumetric flux q between two network nodes is given by

the Hagen-Poiseuille equation:

$$q = \frac{w^3}{12\mu_f} \frac{\Delta P}{L_p} \quad (3)$$

where w is the channel aperture width, L_p is the length of the narrow channel defined as the sum of the radii of the two adjoining particles, ΔP is the pressure difference across the channel, and μ_f is the dynamic viscosity of the fluid.

In each domain, the change in fluid pressure ΔP over a timestep Δt is obtained from the continuity equation:

$$\Delta P = \frac{K_f}{V_c} \left(\sum q \Delta t - \Delta V_{r,\text{fluid}} \right) \quad (4)$$

where $\sum q$ is the total inflow from surrounding channels

into the domain, K_f is the bulk modulus of the fluid (controlling fluid compressibility), V_c is the volume of the domain, and $\Delta V_{r,\text{fluid}}$ is the volume change of the fluid participating in the reaction.

Evidence shows that rock porosity and permeability decrease with increasing confining pressure (Aschwanden et al., 2019). Therefore, when the contact force is compressive, the aperture is defined as (Zhang et al., 2013)

$$w = \frac{w_0 F_0}{F + F_0} \quad (5)$$

where w_0 is the initial aperture of the flow channel, F is the current compressive normal force at the contact, and F_0 is the normal force at which $w = 0.5w_0$.

Table 1: Input parameters for the numerical model.

Symbol [Unit]	Description	Value
<i>Hydraulic parameters</i>		
q_{f0} [m^3/s]	Injection flux rate	3.0×10^{-3}
K_f [MPa]	Fluid bulk modulus	1.57
μ_f [Pa·s]	Fluid viscosity	1.35×10^{-3}
F_0 [N]	Reference force	1.0×10^5
w_0 [m]	Initial aperture	2.0×10^{-7}
<i>Reaction parameters</i>		
k_c [s^{-1}]	Reaction rate	30
β [–]	Coefficient of dilation (2D)	0.27
v_W [L/mol]	Molar volume of fluid	18.1×10^{-3}
v_A [L/mol]	Molar volume of solid A	43.7×10^{-3}
v_B [L/mol]	Molar volume of solid B	133.13×10^{-3}
c_0 [mol/L]	Initial molar concentration	22.84
r [–]	Stoichiometric coefficient r	1.5
s [–]	Stoichiometric coefficient s	0.5
<i>Contact parameters</i>		
E_{contact} [GPa]	Particle contact modulus	80
K_n/K_s [–]	Contact normal-to-shear stiffness ratio	3.5
μ [–]	Particle friction coefficient	0.5
$\bar{E}_{pb}$ [GPa]	Parallel bond (PB) modulus	80
$\bar{K}_n/\bar{K}_s$ [–]	PB normal-to-shear stiffness ratio	3.5
$\bar{\sigma}_c$ [MPa]	PB tensile strength	90
$\bar{c}$ [MPa]	PB cohesion	450
$\bar{\phi}$ [°]	PB friction angle	18

If the normal contact force is tensile or the contact bond is broken, the aperture is defined by the gap between two particles (Li et al., 2024)

$$w = \min(w_0 + \lambda w_m, w_{\max}) \quad (6)$$

where w_m is the gap between the two particles, $w_{\max}$ is an upper bound introduced to ensure computational efficiency, and λ is a dimensionless coefficient.

The aperture also evolves due to precipitation of new minerals. Assuming a two-dimensional geometry with unit thickness in the out-of-plane direction, the aperture change can be quantified as

$$\Delta w = \sum \frac{-(s v_B - v_A) k_c c V_{r,\text{matrix}} \Delta t}{L_p} \quad (7)$$

where L_p is the channel length. The minimum aperture is

set to $w_{\min} = 0.01w_0$ to ensure numerical stability.

The associated fluid volume change $\Delta V_{r,\text{fluid}}$ is computed using first-order hydration kinetics as

$$\Delta V_{r,\text{fluid}} = -r\nu_W k_c c V_{r,\text{matrix}} \Delta t \quad (8)$$

where ν_W is the molar volume of the fluid, k_c is the reaction rate, c is the concentration of solid A, and $V_{r,\text{matrix}}$ is the matrix volume.

Assuming isotropic volume expansion, the particle radius evolves as

$$R = R_0 \left(1 + \beta \frac{c_0 - c}{c_0} \right) \quad (9)$$

where R_0 is the initial particle radius, c_0 is the initial concentration of solid A, and β is a dimensionless swelling coefficient.

Uniaxial compression and direct tensile tests were simulated to calibrate the contact parameters of olivine. Figure 2 shows the results of the uniaxial compression and uniaxial tensile tests. The fracture toughness K_{Ic} can be approximated using the following equation (Huang et al., 2019):

$$K_{Ic} = (2.15\sigma_t + 1.335\sigma_h)\sqrt{R} \quad (10)$$

where σ_t is the tensile strength of the contact bond, σ_h is the minimum principal stress of the confinement, and R is the particle's radius.

From Eq. (10), the estimated fracture toughness K_{Ic} is approximately $3.1 - 3.5 \text{ MPa} \cdot \text{m}^{0.5}$, which is within a reasonable range for typical rocks. Consequently, the fracture process in this scaled sample is representative of larger samples.

3 RESULTS AND DISCUSSION

3.1. Non-reactive Fluid Injection

To replicate field operations, the numerical model applies an injection–shut-in–re injection protocol including three stages: (1) fluid injection, (2) shut-in, and (3) re injection. Fig. 3a shows the pore-pressure evolution for two tests with identical shut-in duration but triggered at different times, while Fig. 3b presents the corresponding injection-rate histories. For clarity, both pore pressure and time are normalized to facilitate comparison. During the injection stage, pore pressure rises rapidly to a peak due to the initially low permeability. The subsequent pressure decline

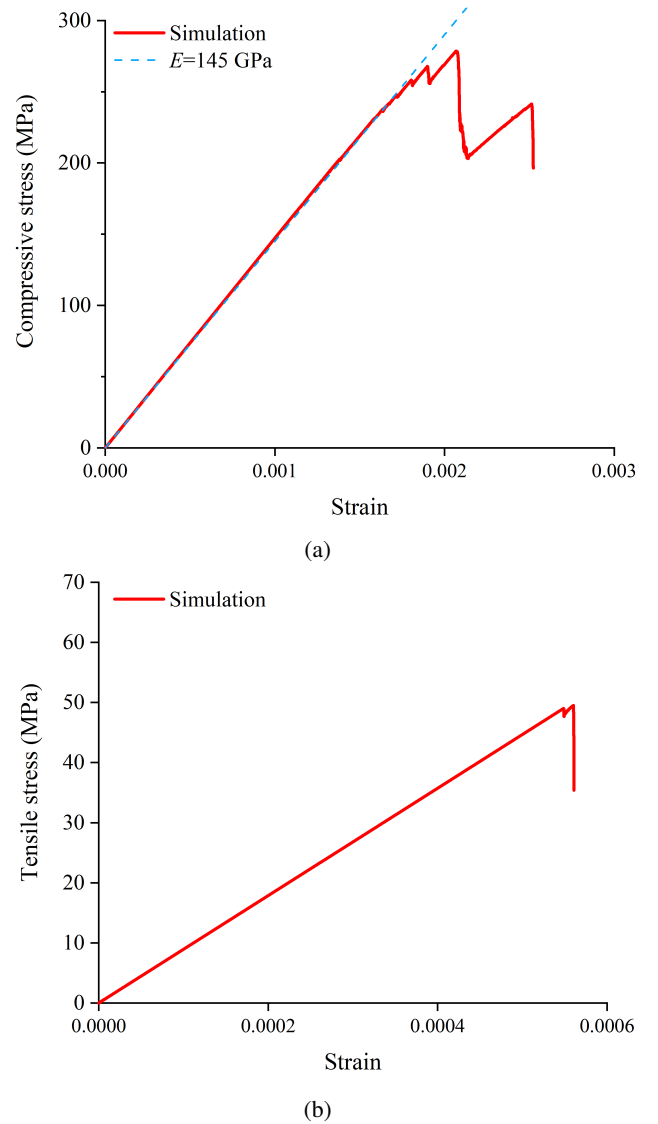


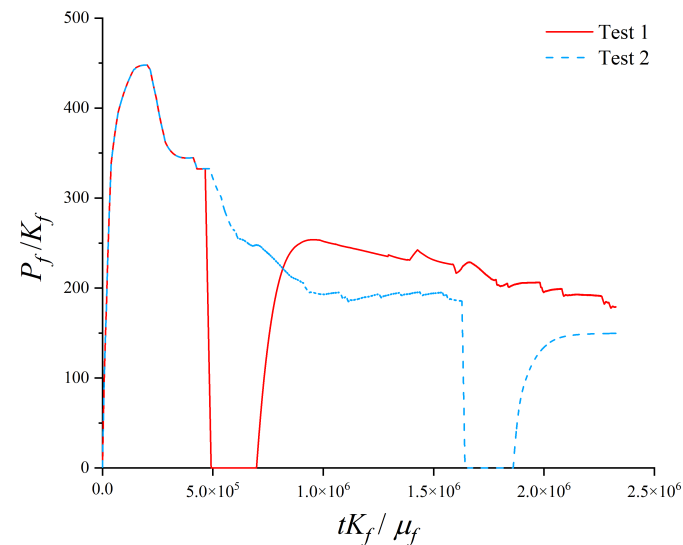
Fig. 2: Calibration of the contact parameters for peridotite (olivine): (a) uniaxial compression test; (b) uniaxial tensile test.

indicates permeability enhancement associated with fracture initiation and growth. During shut-in, the injection rate is set to a negative value to accelerate pressure dissipation. In the re injection stage, pressure increases again as constant-rate injection resumes and approaches a new quasi-steady level. Test 2 exhibits a lower reopening pressure than Test 1, suggesting slightly higher effective injectivity, although both tests follow the same sequence of pressurization, relaxation, depressurization, and recovery. This difference occurs because, while both tests share the same shut-in duration and target extraction flux, a pressure cut-off is enforced to prevent unphysical negative pore pressures. Consequently, because Test 1 enters the shut-in phase at a higher pressure, it sustains the active extraction rate longer before hitting the cut-off boundary, thereby extracting a greater total volume of fluid than Test 2 and creating a higher pressure threshold for subsequent fracture

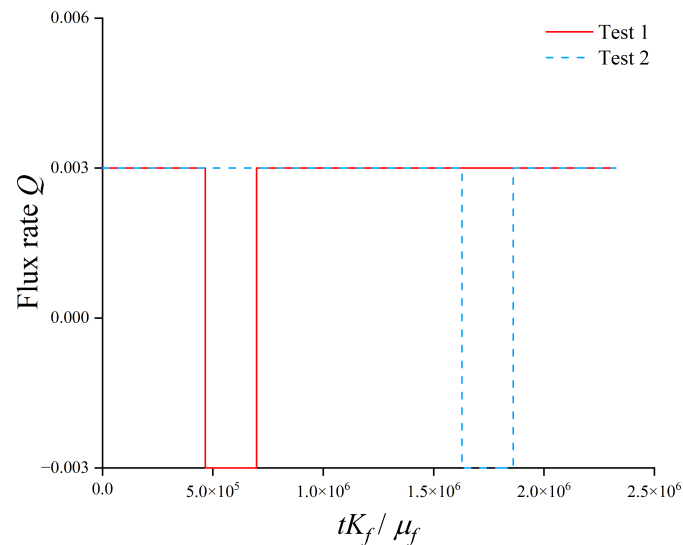
reopening.

3.2. Comparison between Non-reactive and Reactive Fluid Injection

Fig. 4 shows the evolution of normalized pore pressure versus normalized time under an identical injection sequence for non-reactive and reactive fluid injections. In both cases, pore pressure rises rapidly during constant-rate injection and subsequently declines toward a lower-pressure regime as progressive damage enhances permeability. Compared with the non-reactive case, reactive fluid injection produces a distinct second pressure peak and more pronounced fluctuations. This behavior indicates a highly intermittent evolution of hydraulic resistance, primarily driven by reaction-induced clogging.



(a)



(b)

Fig. 3: Simulations characterized by identical shut-in durations but different shut-in start times: (a) pore pressure; (b) injection rate.

After injection resumes, the two cases also diverge markedly in their pressure responses. The non-reactive case approaches a smooth, quasi-steady pressure that gradually decreases over time, consistent with a largely monotonic permeability increase dominated by pure hydraulic fracturing. In contrast, the reactive case exhibits elevated peak pressures and strong post-reopening oscillations, reflecting reaction-driven, time-dependent alterations in flow resistance. These transient pressure spikes indicate the repeated formation and collapse of localized flow pathways as mineral reactions progressively modify pore throats and fracture surfaces. As shown in Fig. 5, reactive injection yields a longer, more branched crack network and a wider damage zone than the non-reactive case. This morphological difference demonstrates that chemo-mechanical feedbacks promote more spatially distributed and temporally intermittent fracture growth. Fig. 6 shows the fluid pressure distribution during both non-reactive and reactive fluid injections. The results align closely with the observed fracture patterns. In the reactive case, mineral precipitation occurs locally, causing aperture reduction and fluid consumption at the pore scale. Consequently, regions exhibiting the strongest pressure localization in Fig. 6b directly correspond to zones where precipitation-induced clogging is most pronounced. These clogged zones are concentrated near the borehole and along the flow paths, thereby promoting the onset of fracture branching and the broader damage zone shown in Fig. 5b.

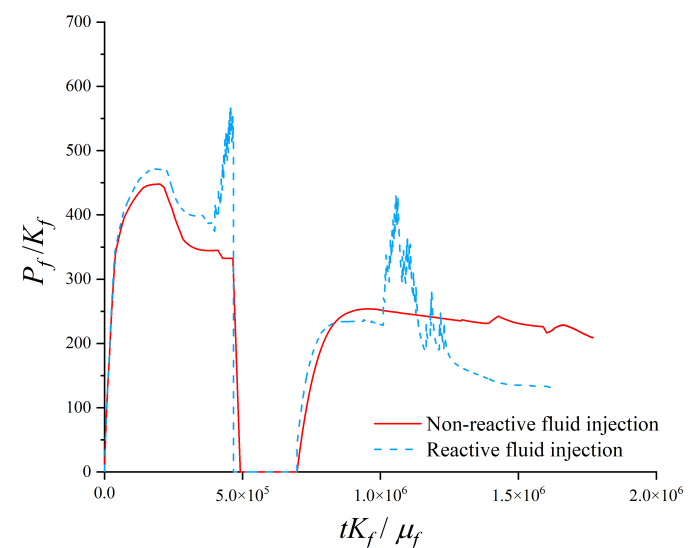
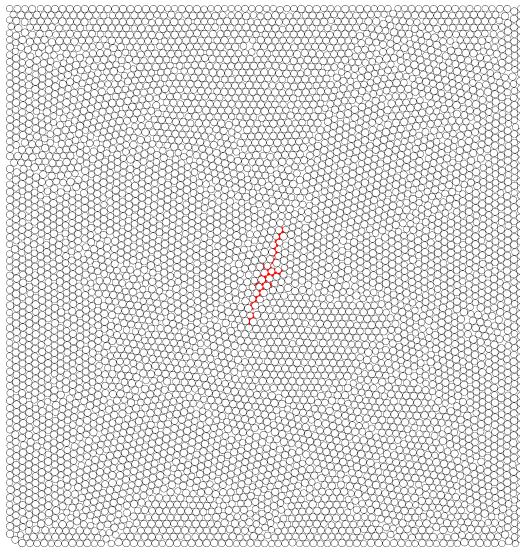
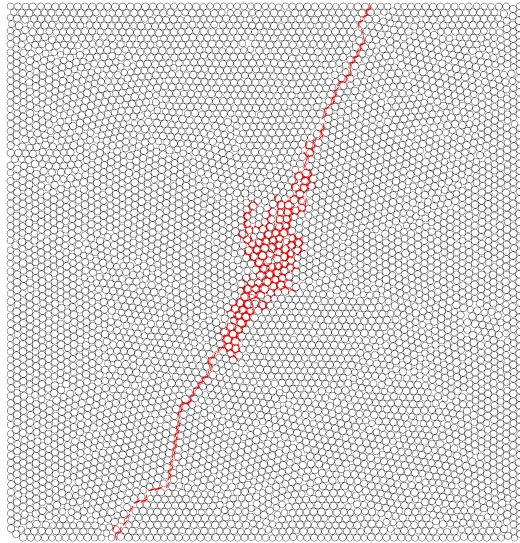


Fig. 4: Pore pressure evolution for simulated experiments with non-reactive and reactive fluid injection.



(a)

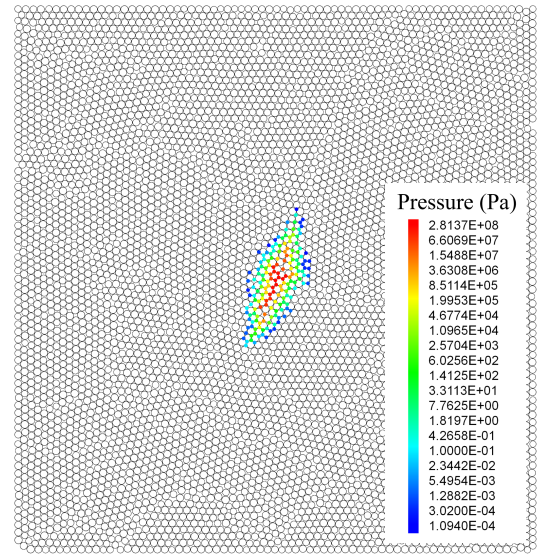


(b)

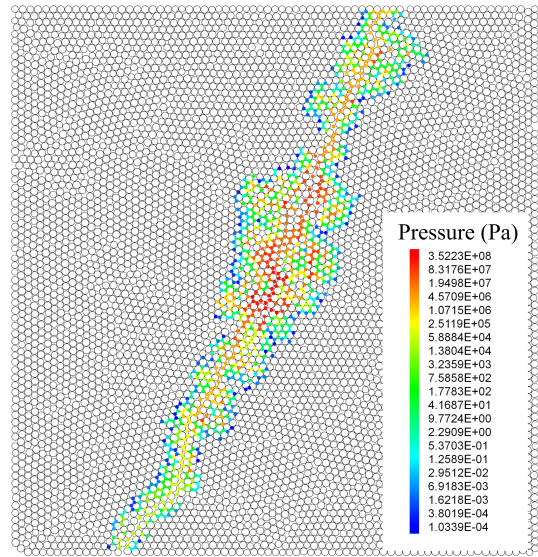
Fig. 5: Fracture pattern for (a) non-reactive and (b) reactive fluid injections. The red line indicates the fracture sites.

4 CONCLUSIONS

A coupled reactive-transport-deformation model defined at the scale of rock pores and particles is used here to simulate an injection-reopening sequence in a representative ultramafic rock reservoir setting. Under constant flow-rate injection, the normalized pressure P_f/K_f rises rapidly at early stages before decreasing as fracturing enhances hydraulic connectivity. For non-reactive injection, varying the shut-in duration primarily shifts the timing of pressure dissipation and recovery, while the reopening pressure provides an inverse proxy for injectivity under a fixed injection rate. In contrast, reactive injection generates significantly higher transient peak pressures and pronounced pressure fluctuations, reflecting a highly intermittent evo-



(a)



(b)

Fig. 6: The pressure distribution for (a) non-reactive and (b) reactive fluid injections.

lution of hydraulic resistance. Morphologically, reactive injection promotes a longer, more branched fracture network and a broader damage zone. Ultimately, these findings underscore the critical role of coupled chemo-hydro-mechanical feedback in governing fracture development and overall injectivity during in-situ mineralization.

ACKNOWLEDGMENTS

This work was supported as part of the Center on Geoprocesses in Mineral Carbon Storage, an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Science, Basic Energy Sciences at the University of Minnesota under award DE-SC0023429.

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